MED 2045. Quantitative Methods.
In this course, students will be introduced to fundamental concepts in clinical epidemiology and basic statistics, as they relate to population and clinical research. This course is intended to teach students both the basic knowledge required to develop and interpret clinical studies as well as the skills in order to conduct basic statistical analyses.

MED 2046. Leadership in Health Care.
This courses emphasizes practical application of teamwork and leadership skills across multiple settings. Leadership in Health Care is a master's level course for second year medical students enrolled in the Primary Care-Population Medicine (PC-PM) program. Through interactive classroom sessions, field work in health care advocacy, and a team-based "leadership action project", students will develop foundational leadership skills. The first formal leadership course at Alpert Medical School, Leadership in Health Care will contribute to the PC-PM program's ultimate goal of preparing physician leaders who will improve the quality of health care and wellness of the population.

MED 2050. HSS III: Pop & Clinical Med I.
This is the first semester of Population and Clinical Medicine, a two-semester course focused on the integration of population medicine and clinical practice. In this course, students will focus on topics integral to clinical medicine, but expand beyond the patient into the population and beyond. Given the importance of population health interventions for impacting the health of vulnerable and underserved patients, the course will focus on issues affecting these populations.

MED 2060. HSS III: Pop & Clinical Med II.
This is the second semester of Population and Clinical Medicine, a two-semester course focused on the integration of population medicine and clinical practice. In this course, students will focus on topics integral to clinical medicine, but expand beyond the patient into the population and beyond. Given the importance of population health interventions for impacting the health of vulnerable and underserved patients, the course will focus on issues affecting these populations.

MED 2070. Health Systems Science IV.
This course is designed to further explore the themes of the Primary Care-Population Medicine Program and prepare students for the next steps in their professional careers. The course is designed to be a capstone and employs integrated, developmental, evolutionary educational spirals - providing the knowledge, attitudes and skills at the right time in the right format, and building on the first three years of the program. Course threads include Health Systems Science Advanced Content, Skill Building, Preparation for Next Career Stages, and Master's Thesis Workshopping.

MED 2110. Introduction to Medical Sciences and Patient Care.
This 2-week intensive course introduces students to the wide variety of topics explored in the Master's of Medical Sciences program, with a focus on patient care aspects. The course combines seminar classroom instruction with field work/immersion at community healthcare sites. Topics covered include: biopsychosocial model of healthcare; intersection between science, social science and humanities in healthcare; introduction to community health centers; professionalism in healthcare; basic healthcare communication skills; quality improvement skills; and strategies for mastery of basic science knowledge. Students will be assessed using multiple methods including: seminar participation, reflective essays/field notes, attendance at field-work sites, & assessment from community mentors.

MED 2120. Patient Care in Complex Systems I.
This is the second of a three course series for Master of Medical Sciences students. This course introduces students to the variety of complex factors affecting health, imparting both theoretical knowledge and practical skills. Teaching methods: interactive seminars and experiential learning at community healthcare sites with members of multidisciplinary teams. Topics covered: healthcare systems, social determinants of health, roles of interdisciplinary healthcare team members, quality improvement, and epidemiology. Students will begin developing a project at their clinical sites which will be implemented in spring semester. Student assessment includes: seminar participation, reflective essays, attendance at field-work sites, and assessment from community mentors. Pre Requisites: MED 2110

MED 2130. Patient Care in Complex Systems II.
This is the third of a 3 course sequence for Master of Medical Science students. Students will continue their study of both theoretical and practical aspects of healthcare through an interactive seminar series, continued service learning at their longitudinal community healthcare site, and completion of their community project. Seminar topics: care of vulnerable populations, environmental health, population health, new models of healthcare delivery, ethical issues in healthcare, whole-person health, cultural humility, complementary and alternative medicine, and patient advocacy. Students assessment includes: seminar participation, reflective essays, attendance at field work sites, assessment from community mentors, and quality of project and presentation. Pre Requisites: MED 2110 and MED 2120.

MED 2140. Human Histology.
Human Histology provides an in-depth examination of the basic architecture of the body. Fundamental to this understanding is the cell and how during early development cells in the aggregate undergo specialization as tissues, which are the building blocks of the body. This course focuses first on the biology of the four basic tissues (epithelium, connective tissue, muscle and nerve) and second, how they contribute to the functional anatomy of all organs and systems. We will emphasize characteristic developmental, structure-function and regulatory relationships within normal cells and tissues, which in turn are the foundation for the understanding of pathological alteration.

MED 2150. General Pathology.
Pathology is the study of the causes, mechanisms, and consequences of disease. In General Pathology students study in detail the cell and tissue alterations that lead to the production of human diseases. To uncover such alterations, morphological observations are correlated with studies involving molecular biology, biochemistry, and genetics. In studying the pathogenesis of human disease we pay close attention to epidemiological parameters, population health, aging, and to environmental and occupational health problems. General Pathology been integrated, whenever possible, with other courses in the Fall Semester of the Gateways Program, in order to maximize learning opportunities.

MED 2160. Human Anatomy 1.
This course explores the anatomical organization of the human body, viewing anatomical structures as a product of development and functional demand. Human Anatomy provides an opportunity for students of diverse backgrounds, interests, and goals to emerge with an understanding of the human body as a cornerstone of medical science. The course uses a combination of lectures, on-line modules, and mandatory laboratory sessions examining human cadaver prosections, to impart broad conceptual and in-depth knowledge of this subject.

MED 2170. Scientific Foundations of Medicine.
Scientific Foundations of Medicine is an integrated cross-disciplinary course that introduces the fundamental basic science principles relevant to the study of health, disease mechanisms and clinical medicine. As such the course consists of six blocks of core topics that incorporate foundational principles of molecular biology, cellular and metabolic biochemistry, nutritional science, cell physiology, inheritance patterns, mechanisms of genetic disorders, and immunology. Grounding in these scientific principles gives students insight into the biological complexity and genetic diversity that underlies disease processes.

MED 2180. Brain Sciences and Neurological Disease.
Brain Sciences is composed of several interrelated components - Head Anatomy, Neurobiology, Neuropathophysiology, Neuropathology and Neuropharmacology. The intent of the course is to encourage the integration of underlying neuroanatomy and basic science principles with an understanding of the presentation and management of neurological diseases. Course leaders from each of these disciplines have worked closely together in order to present the material in a cohesive and logical framework that promotes the sequential acquisition of new information based upon a substantive understanding of the previous material.